NITHISHVELAN M

Game Systems & Engine Programmer
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PROFESSIONAL SUMMARY

Game Systems & Low-Level Programmer specializing in modern C++17/20, Data-Oriented Design (DoD), cache-friendly memory architectures (Sparse Sets), real-time 2D physics simulation, and graphics compositing. Open-source contributor to Valve's Linux gaming infrastructure (`gamescope`, `Proton`) and PlayStation 5 translation tooling (`AnyPS5`). Experienced in designing deterministic game loops, SAT collision solvers, and low-latency rendering pipelines.

CORE TECHNICAL SKILLS

- **Languages & Standards:** C++17, C++20, C, Python 3, POSIX Shell, CMake.
- **Game Engine Architecture:** Data-Oriented Design (DoD), Entity Component System (ECS), Sparse Sets, Cache Line Saturation, Fixed-Timestep Accumulators, State Machines (FSM).
- **Physics & Math:** Separating Axis Theorem (SAT), Impulse-Based Collision Response, Minimum Translation Vectors (MTV), A* Pathfinding (Manhattan/Octile), Procedural Dungeon Generation (BSP).
- **Graphics & Systems:** Vulkan 1.3, PipeWire, POSIX Sockets, Multi-threading, AddressSanitizer (ASan), UndefinedBehaviorSanitizer (UBSan), Valgrind.

UPSTREAM OPEN SOURCE GAME & COMPATIBILITY CONTRIBUTIONS

- **ValveSoftware/gamescope** — *Linux Gaming Session Compositor & Vulkan Pipeline*
- Diagnosed and resolved video corruption during Steam Remote Play sessions on NVIDIA proprietary drivers by introducing multi-planar capability checks on NV12 storage images (`VK_IMAGE_USAGE_STORAGE_BIT`).
- Addressed DRM format modifier stride alignment issues when exchanging video frames across PipeWire and NVENC hardware encoders.
- **ValveSoftware/Proton** — *Steam Play Windows Compatibility Tool (Wine)*
- Resolved Linux case-sensitivity collisions during Windows DLL resolution by correcting DriverStore directory path lookups and file descriptor lifecycle management.
- **boykopovar/AnyPS5** — *PlayStation 5 Systems PRX Translation Layer*
- Implemented `sceSysmoduleIsLoadedInternal` lifecycle state tracking (*merged upstream*), implemented FreeBSD `tgammaf` precision routines, and audited PRX exports against upstream Python AST header integrity tests.

FEATURED GAME ENGINE & SYSTEMS PROJECTS

game-dev-labs — *Data-Oriented Game Engine Architecture Suite (C++17)*

- **Sparse Set Entity Component System (ECS):**
- Built an ECS utilizing the **Sparse Set** data structure, guaranteeing $O(1)$ component addition, lookup, and swap-and-pop deletion while maintaining 100% cache-line saturation.
- Verified $O(1)$ defragmentation with zero virtual dynamic dispatch overhead in simulation loops; processes **50,000 entities in `< 0.82ms`**.
- Authored **ADR 0001: Sparse Sets vs. Contiguous Dense Arrays in ECS**.
- **2D Physics Engine (SAT Rigid Body Dynamics):**
- Implemented the **Separating Axis Theorem (SAT)** for arbitrary convex polygons and circles, extracting Minimum Translation Vectors (MTV) for contact manifolds.
- Engineered an impulse-based momentum and angular rotation solver incorporating Coulomb friction and positional slop projection to prevent object sinking.
- **Procedural Generation & A* Pathfinding:**
- Implemented Binary Space Partitioning (BSP) dungeon generation with 100% topological room connectivity and 4-5 Cellular Automata cave smoothing.
- Built an optimized A^* navigation solver with Octile/Manhattan distance heuristics and priority queue tie-breaking.
- **Deterministic Game Loop & FSM:**
- Implemented Glenn Fiedler's fixed-timestep accumulator with state interpolation alpha (α) to decouple physics execution from variable refresh rates.

AnyPS5-Launcher — *Modern Desktop Launcher & Reverse Engineering Hub*

- Developed a desktop launcher and executable analyzer for PS5 homebrew/relinked binaries, providing automated ELF header validation, PRX dependency resolution, and runtime controller mapping.

ENGINEERING DISCIPLINE & RELIABILITY

- **Sanitizer-Driven CI:** All C++ simulation code is validated on GitHub Actions across **Ubuntu and Windows** under **AddressSanitizer (ASan)** and **UndefinedBehaviorSanitizer (UBSan)**.
- **Memory Integrity:** Validated zero memory leaks using Valgrind Memcheck (`0 bytes lost in 0 blocks`).

EDUCATION

- **Bachelor of Engineering (B.E.) in Materials Science and Engineering**
  College of Engineering, Guindy (CEG), Anna University, Chennai